

Lecture Material #7

Information Security

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Quantum Attack





Quantum Computing

Key Principles of Quantum Computing

- **Qubits (Quantum Bits):** Unlike traditional binary bits (0 or 1), a qubit can be 0, 1, or a combination of both simultaneously, allowing for massive parallel processing.
- **Superposition:** Qubits can exist in a "spinning" state, acting as both 0 and 1 at the same time until measured.
- **Entanglement:** Qubits can be linked, meaning the state of one qubit instantaneously depends on another, even over distances, allowing for exponential increases in computational power.
- **Interference:** Quantum algorithms use interference to cancel out incorrect paths and amplify the correct answer.



Quantum Computing

How Does Quantum Computing Works?

Qubits



Classical Bit : 1
Qubit : 0 & 1 at
the same time

Superposition



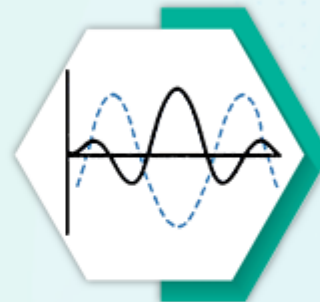
A qubit can be
in a combination
of 0 and 1

Entanglement



Qubits can be
correlated with
each other

Quantum Interference



Probability
amplitudes of
qubit states
can interfere

Quantum Parallelism



A quantum
computer can
evaluate multiple
inputs at once





Quantum Computing

Bit
0



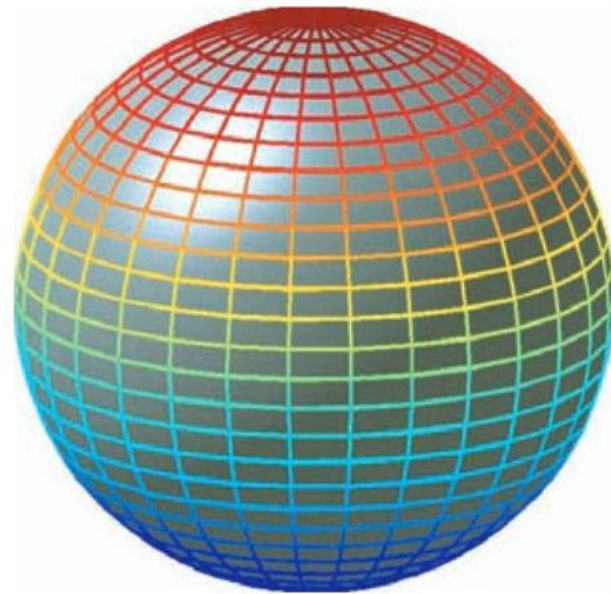
1

Pbit
0



1

Qubit
 $|0\rangle$



$|1\rangle$





Quantum Computing



Quantum Attack

◆ Why Qubits Are Powerful

Because of superposition + entanglement:

- 1 qubit → represents 2 states
- 2 qubits → 4 states
- n qubits → 2^n states

👉 This allows quantum computers to:

- Solve certain problems much faster than classical computers
- Work efficiently on cryptography, optimization, simulations, etc.



Quantum Computing

How It Differs from Classical Computing

- **Processing Speed:** While a classical computer checks possibilities one by one, a quantum computer can evaluate many possibilities simultaneously.
- **Scaling:** Every added qubit exponentially increases processing capacity, unlike the linear growth of traditional transistors.
- **Measurement:** When measured, the superposition collapses into a definite 0 or 1, providing the final result.



Quantum Attack

A **quantum attack** is a cryptographic attack that uses **quantum computers** and quantum algorithms to break or weaken classical cryptographic systems.

Why It Matters

- Quantum computers can solve certain mathematical problems **much faster** than classical computers.
- Many widely used cryptographic schemes rely on problems that are **hard for classical computers** but easy for quantum ones.

Key Quantum Algorithms

- **Shor's Algorithm** → breaks public-key cryptography
- **Grover's Algorithm** → weakens symmetric cryptography



Quantum Attack

Classical vs Quantum Computers

Classical Computers

- Limited by exponential or sub-exponential complexity
- RSA/ECC considered secure under classical assumptions

Quantum Computers

- Exploit quantum parallelism and interference
- Can solve factoring and discrete logarithms in **polynomial time**



Quantum Attack



Why RSA is Vulnerable to Quantum Attacks

RSA Security Depends On

- Difficulty of integer factorization
- Example: factoring large numbers (2048-bit keys)

Quantum Impact

- **Shor's Algorithm** can factor large integers efficiently
- RSA becomes **completely breakable** once large-scale quantum computers exist



Quantum Attack



Why ECC is Vulnerable to Quantum Attacks

ECC Security Depends On

- Elliptic Curve Discrete Logarithm Problem (ECDLP)

Quantum Impact

- Shor's Algorithm also solves discrete logarithms efficiently
- ECC loses its security advantage despite smaller key sizes



Quantum Attack



Why Symmetric Algorithms Are Less Vulnerable

Examples

- AES
- DES (already weak, not quantum-related)
- Hash functions (SHA-256, SHA-3)

Quantum Impact

- Grover's Algorithm provides only a quadratic speedup
- Security loss is manageable by increasing key size

Example

- AES-128 → equivalent to ~64-bit security
- AES-256 → still considered quantum-resistant





IPSec (IP Security) Architecture





IPSec (IP Security) Architecture

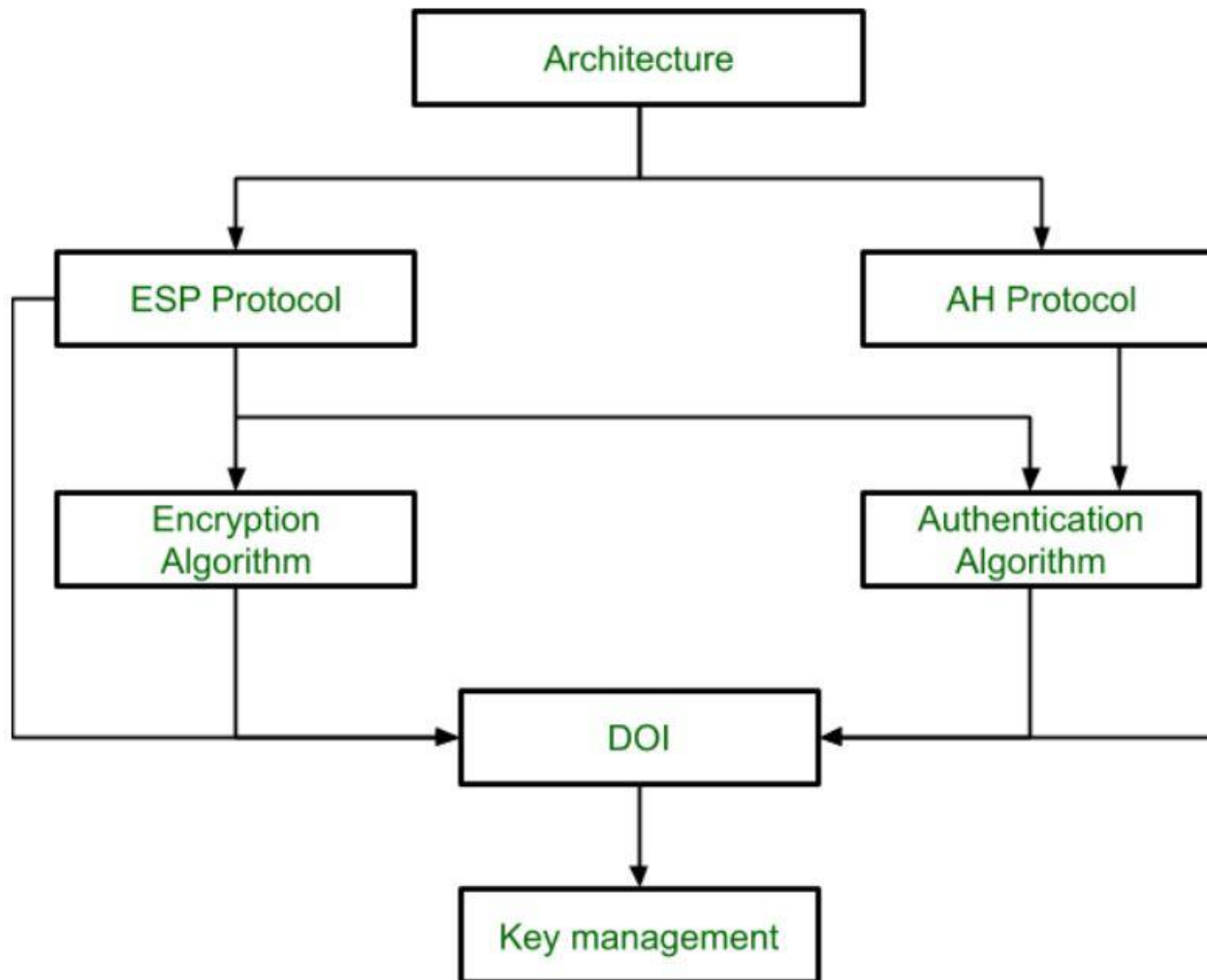
IPSec (IP Security) architecture uses two protocols to secure the traffic or data flow. These protocols are **ESP** (Encapsulation Security Payload) and **AH** (Authentication Header). IPSec Architecture includes protocols, algorithms, DOI, and Key Management. All these components are very important in order to provide the **three main services**:

- ☐ Confidentiality
- ☐ Authentication
- ☐ Integrity





IPSec (IP Security) Architecture





IPSec (IP Security) Architecture

- 1. Architecture:** Architecture or IP Security Architecture covers the general concepts, definitions, protocols, algorithms, and security requirements of IP Security technology.
- 2. ESP Protocol:** ESP (Encapsulation Security Payload) provides a confidentiality service.
- 3. Encryption algorithm:** The encryption algorithm is the document that describes various encryption algorithms used for Encapsulation Security Payload.
- 4. AH Protocol:** AH (Authentication Header) Protocol provides both Authentication and Integrity services.





IPSec (IP Security) Architecture

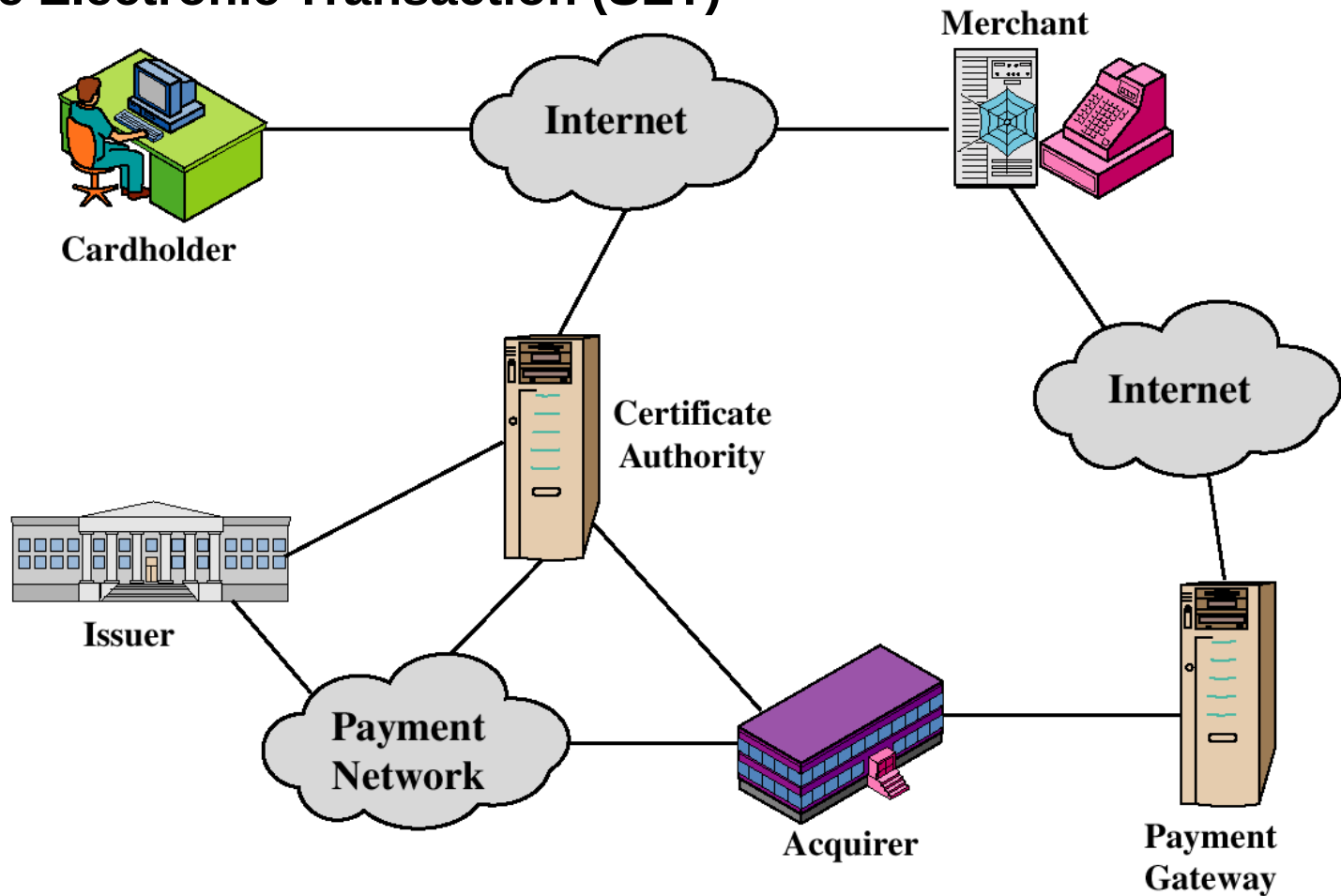
5. **Authentication Algorithm:** The authentication Algorithm contains the set of documents that describe the authentication algorithm used for AH and for the authentication option of ESP.
6. **DOI (Domain of Interpretation):** DOI is the identifier that supports both AH and ESP protocols. It contains values needed for documentation related to each other.
7. **Key Management:** Key Management contains the document that describes how the keys are exchanged between sender and receiver.





Handshaking

Secure Electronic Transaction (SET)





Handshaking



Handshaking

Handshaking is an **automated process of negotiation** that dynamically sets parameters of a **communications channel** established between two entities before normal communication over the channel begins. It follows the physical establishment of the channel and precedes normal information transfer.





Handshaking

Handshaking

